Ayush Chachan

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Education

Indian Institute of Technology, Delhi — July 2016 – June 2022
Dual Degree (B. Tech and M. Tech) in Biochemical Engineering and Biotechnology — *CGPA - 8.905/10.000*

Experience

UPSC Civil Services Preparation | — Oct 2023 – Oct 2025
- Prepared for UPSC CSE with **Mathematics Optional**, involving sustained self-study and structured learning.

Helloverify India Private Limited | *Software Engineer* — July 2022 – Mar 2023
- Built scalable REST APIs using **Django REST Framework**, improving client onboarding speed and efficiency.
- Engineered **Selenium**-based web scrapers to automate data extraction, reducing the time by **82% per request**.
- Integrated **third-party APIs** to enrich verification workflows, enhancing overall system capability.
- Implemented **Celery**-based **schedulers** for asynchronous email and phone alerts, ensuring timely client updates.

Indian Institute of Technology, Delhi | *Graduate Researcher (M.Tech Thesis)* — May 2021 – June 2022
- Designed **statistical algorithms** for bioreactor data validation using mass balances and regression models.
- Applied **error propagation analysis** and consistency testing to identify anomalous and unreliable process data.
- Developed Python routines for **least squares estimation** and **chi-square testing** in over-determined systems.
- Utilized **statistical modelling methods** to improve data reliability and automate process control decisions.

Scholastic Achievements

IIT Delhi BOSS Award (2022) - Conferred for best experimental and design-based research in the **M.Tech project**.

Projects

Large-Scale Funnel Analytics Optimization | *Java, Binary Protocol*
- Re-architected a self-join–intensive analytics query into a user-level aggregation model, reducing algorithmic complexity from **$O(n^2)$** to **$O(n)$** and lowering BigQuery slot consumption by **80%**.
- Developed scalable **BigQuery JavaScript UDFs** operating on aggregated user event arrays, applying **binary-search–based** evaluation for efficient sequential funnel processing at scale.
- Built a fixed-schema analytics ingestion pipeline using **Protobuf**, reducing network payload size by **40%** and improving serialization and deserialization throughput by **2×**.
- Containerized analytics services with **Docker** and automated deployments using **Kubernetes** and **GitLab CI/CD** for reliable and repeatable production releases.

Unix-like Shell Implementation | *C, UNIX System Calls*
- Developed a shell using **system calls (fork, exec, wait)**, with **zombie process reaping** and error handling.
- Implemented **background(&), serial(&&) and parallel(&&&)** execution and custom **signal handlers**.

Monte Carlo Simulation | *Disjoint Sets, Union Find, Java*
- Modeled 2-D lattices using **Graph** and estimated the percolation threshold using Monte Carlo Simulation.
- Improved the running time of simulation using **Disjoint Set** data structure and the **Union-Find** algorithm.

K-Dimensional Tree | *Java, BST, K-Nearest Neighbour search*
- A *K-D Tree* is a space-partitioning data structure for efficiently indexing points in a **K-Dimensional space**.
- Implemented *2d-tree* with Insert, Search, **Range Search** and **Nearest Neighbour** search in $O(\log_2 n)$ time.

Integer Arithmetic and Irrationals | *Karatsuba Multiplication, Newton's Method*
- Implemented **High Precision Integer Multiplication** and **Division** using the **Karatsuba's** algorithm.
- Used **Newton's Method** to compute **irrationals up to one million digits** precision in $O(\log_2 d)$ time..

Technical Skills

Programming Languages: Java, Python, SQL, Java Script, C/C++, Bash (Shell Scripting).
Frameworks & Libraries: Spring Boot, Hibernate/JPA, Django REST Framework, Celery
Databases & Storage: MySQL, PostgreSQL, Redis.
Web & API Development: REST APIs, Microservices, JSON, HTML5, CSS3.
Developer Tools: Git, Maven, Amazon Q, Bash, .
Data Analysis: Jupyter, NumPy, Pandas, Excel (Data Visualization).